Team 9 – Final Report and Instruction Manual

EGR 201: Interactive Teaching and Learning Tool for Quartz Hill Elementary
Team Members: Anthony Buvelot, George Williams, Tessa Cummings, Okenna Anthony-Wise

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Executive Summary

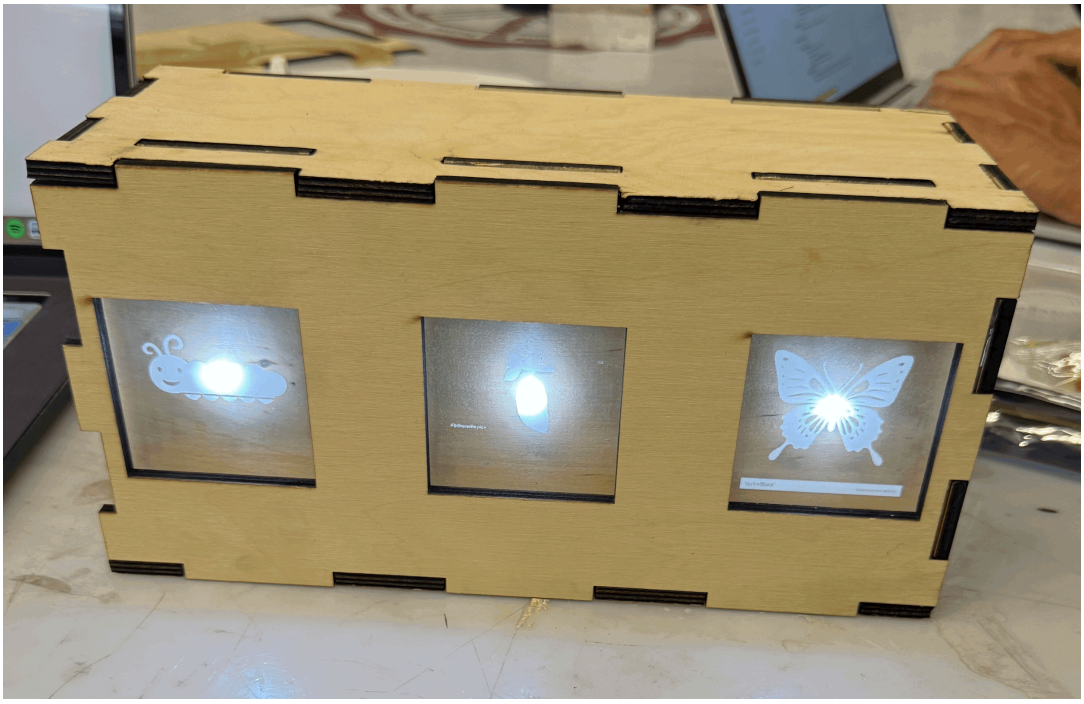


Figure 1: Evolution Showcase Device

The purpose of our device is to educate elementary students about evolution and the adaptations animals undergo to survive in harsh environments. Our goal was to create a hands-on, interactive, and informative tool that fosters engagement. We developed an evolution showcase device that allows students to explore various animal adaptations through interchangeable slides.

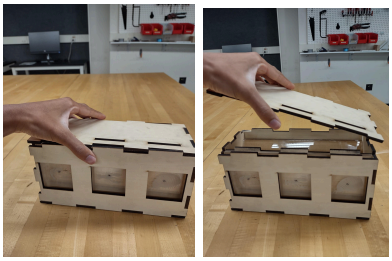
Key constraints included a budget of $200 and the integration of mechanical components such as coding, 3D printing, SolidWorks modeling, and laser cutting. Additionally, the design needed to be safe, simple, age-appropriate, durable, and easily stored in a classroom setting. To meet

these criteria, we created a compact box (Figure 1) with fewer components, enhancing both usability and storage efficiency.

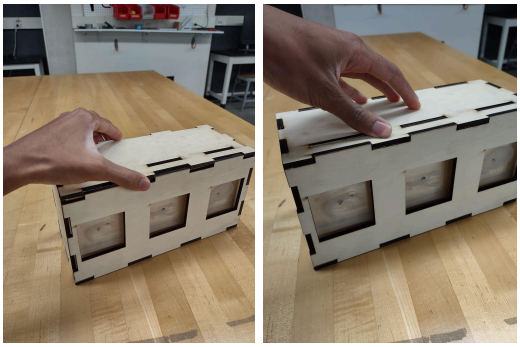
User's Guide

1. Assembly Instructions

- **Step-by-Step Guide:**
    - Firmly grasp the lid and remove it.



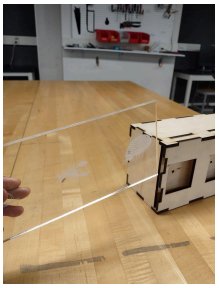
    - Align the lid with the opening and reinsert it securely.



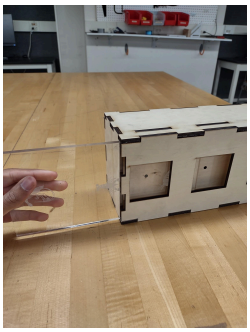
- **Required Tools and Materials:**
    - No tools required.
- **Additional Support:**
  Diagrams or images can be included as needed to illustrate the assembly process.

2. Operating Instructions

- **Procedure:**
    1. Locate the desired slide.



2. Insert the slide into the designated cubby.



3. Remove the lid and insert the battery into the compartment.
4. Press the switch button to turn on the light.
5. To light up all the boxes, press the switch button four times.
6. To turn off the device, press the switch button five times.

3. Troubleshooting Tips

- **Common Issues and Solutions:**
  - **Issue:** Slides not displaying correctly.
    - **Solution:** Ensure the slides are facing the correct way.
  - **Issue:** Device not powering on.
    - **Solution:** Verify the battery is properly inserted and the lid is fully secured.
- **Maintenance:**
  - Replace batteries every 2–3 months or as needed.

4. Maintenance Guide

- **Storage and Care:**
  - Store in a cool environment (65–78°F).
  - Use 9V batteries.
    - Disconnect Batteries if not in use for long periods of time
  - Dust all parts bi-weekly to keep the device clean.
  - Make sure lights are off when not in use
- **Additional Tips:**

- Don’t mess with wires
- Don’t remove duct tape
- Wood may cause injury
- Electrical part may cause shock
- Handle with care
- Don’t stare at the lights for long periods of time

User Feedback

What users liked most:

- The tool is informative, easy to use, and allows children to operate it independently.
- Its lightweight and compact design is classroom-friendly.
- Features like the light sequences and showcasing evolution/adaptation were well-received.

Suggestions for additional features:

- Improved LEDs and more slides.
- A storage compartment for slides.
- A voice box or audio feature to explain images.
- Different colored lights for better visibility of drawings.
- Enhanced materials and neater internal wiring for durability and aesthetics.

Clarity of instructions:

- Generally clear, concise, and easy to follow for first graders.
- One suggestion mentioned simplifying wording further for clarity.

Student engagement:

- Most users agreed the tool effectively engages students by visually demonstrating evolution and connections between animals.
- Concerns included potential loss of interest over time and dependency on slide availability.

Meeting classroom needs:

- Compact and useful, with potential for broader application by adding slides on different topics.
- Suggestions included integrating slides that explain evolution and expanding to other subjects like math.

Recommendations

Design Modifications:

- Use higher-quality LEDs and more durable materials.
- Add a storage compartment for slides.
- Improve internal wiring aesthetics.

Additional Features:

- Include a voice box or sound feature for auditory learning.
- Use colored lights for better visualization.
- Develop modular slides for different topics to expand educational use.

These changes would better usability, functionality, and appeal for both students and teachers.

Bill of Materials

Item	Description	Cost ($)
Arduino	Arduino Uno System	25.00
9V Batteries	Source of Power	9.35
Snap Switch	Power Control Component	9.99
Male-Male Wires	Electrical Connections	4.99
LED Lights	Lights for Visual Display	4.99
¼” Wood	Structure Material	17.50
¼” Acrylic	Design Material	14.00
Resistors	Current Flow Control	1.99
Total		**$87.81**